

FinFET Semiconductor Device Report

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Abstract – Fin field-effect transistors (FinFETs) are a 3D multigate transistor that has revolutionized modern semiconductor devices by extending Moore’s Law beyond the limits of planar CMOS. This report provides an overview of FinFETs, including their history, fundamental operating principles, and wide-ranging applications in digital logic, memory, analog and RF domains. In the present day, dominating industries like Intel, AMD, and Nvidia illustrate how FinFET technology became the backbone of modern-day processors. The market growth of FinFET-based technology is also explored in this report. The limitations of FinFETs are also discussed and its successor transistor known as Gate-All-Around FETs (GAAFETs) are explored, due to the advantages GAAFET architectures offer for sub-5nm technologies.

INTRODUCTION

Semiconductor device scaling has enabled exponential improvements in computing power for decades, but by the early 2000s, traditional planar transistors such as MOSFETs faced severe short-channel effects and leakage as their gate lengths shrank below 30 nm. FinFET technology emerged as a radical solution to overcome these limits by introducing a three-dimensional “fin”-shaped channel wrapped by the gate on multiple sides [1].

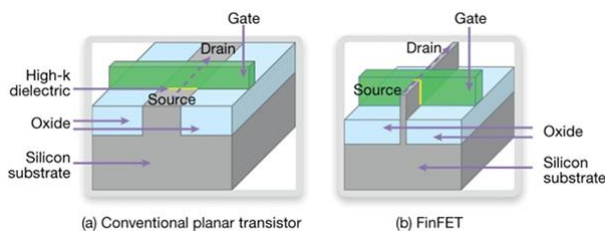


Figure 1: Planar transistor design vs FinFET design [2].

By improving gate control over the channel, FinFETs significantly reduce leakage current and enable faster switching at smaller dimensions than planar MOSFETs [3]. FinFET transistors, sometimes called “Tri-Gate” transistors, were first commercialized in the early 2010s and have since become the dominant transistor architecture for high-performance and mobile chips at 14 nm, 10 nm, 7 nm and 5 nm technology nodes. Nearly all modern CPUs, GPUs, and system-on-chip devices today rely on FinFETs to deliver

superior performance per watt. As one of the cemented transistors in IC technology, this report will explore what makes them so superior to other transistors.

HISTORY

The concept of using multiple gates to better control a transistor’s channel has been proposed since the 1960s. Early work by researchers like H.R. Farrah and R.F. Steinberg (1967) and Toshihiro Sekigawa (1980) proposed double-gate transistor structures to suppress short-channel effects [4] – a result of having channel lengths comparable to the depletion region width of the source and drain junctions leading to non-ideal MOSFET behavior [5]. A significant milestone was achieved in 1989 when Digh Hisamoto and colleagues at Hitachi Central Research introduced the first double-gate MOSFET, called the “Depleted Lean-channel Transistor” (DELTA) [6]. This device had a raised channel (fin) with gates on both sides and demonstrated markedly reduced short-channel effects compared to planar MOSFETs. Throughout the 1990s, various research groups experimented with multi-gate designs to improve scalability. Eventually, in 1996, Effendi Leobandung and Stephen Chou showed that slicing a wide transistor into multiple narrow channels (essentially forming “fins”) could improve current drive – a precursor to the modern FinFET concept. The FinFET as we know it was pioneered at the University of California, Berkeley. In 1999, Prof. Chenming Hu’s team presented a device named the “FinFET,” a self-aligned double-gate MOSFET with a vertical silicon fin as the channel [7].

After a decade in academia, FinFETs made a historic leap to industry in the 2010s after numerous improvements and modifications to its design. Intel was the first company to commercialize FinFETs, announcing in May 2011 that their 22 nm CPU technology (the Ivy Bridge processors) would use 3D “Tri-Gate” transistors instead of planar CMOS [8]. Intel called this the most radical shift in transistor technology in over 50 years, validating FinFET as the new industry standard.

BASIC DEVICE OPERATION

FinFETs operate on the same fundamental principle as conventional MOSFETs. They are typically enhancement-mode transistors that act as voltage-controlled switches, although depletion-mode FinFETs do exist. When a sufficient gate-to-source voltage (V_{GS}) is applied, an inversion layer forms along the fin surfaces, allowing current

to flow between the source and drain. (This is shown by the yellow region in figure 1).

As FinFETs operate on the same fundamental principle as MOSFETs, its modes of operation can be split into 3 regions: the cutoff region, linear region, and saturation region.

Cutoff region ($V_{GS} < V_{TH}$):

When the gate source voltage is below the threshold voltage, no inversion layer is formed. As a result, the source and drain are disconnected, and the only current flowing through arises from very small leakage due to the diffusion of minority carriers.

Linear region ($V_{GS} > V_{TH}$ and $V_{DS} < (V_{GS} - V_{TH})$):

When the gate source voltage is above the threshold voltage and there is a small drain source voltage, the FinFET conducts current through the drain and in this region. Here it varies linearly with V_{DS} . This is also why this region is sometimes referred to as the ohmic region.

Saturation region ($V_{GS} > V_{TH}$ and $V_{DS} \geq (V_{GS} - V_{TH})$):

If gate source voltage is above the threshold voltage and there is a large drain source voltage, the channel pinches off near the drain (for a n-channel FinFET), and at this point the current becomes independent of the drain source voltage and is only controlled by the gate source voltage.

These 3 modes are similar to that of a MOSFET. However due to the design of the FinFET, it is able to provide a greater performance per watt. This can be explained due to the additional surface area provided by the FinFET.

The drain current in a MOSFET is given by the following equations:

1. Linear region:

$$I_D = \mu_n C_i \frac{Z}{L} \left[(V_G - V_T) V_D - \frac{V_D^2}{2} \right]$$

2. Saturation region:

$$I_D = \mu_n C_i \frac{Z}{2L} (V_G - V_T)^2$$

The equations for a FinFET are very similar, with the key difference being the gate width given by Z_{eff} , where $Z_{eff} = 2H_{fin} + Z_{fin}$, where H and W are the height and width of the fin respectively. Similarly, there the oxide capacitance used is typically an adjusted effective value. Taking these factors into consideration the equations for the FinFETs become:

1. Linear region:

$$I_D = \mu_n C_i \frac{Z_{eff}}{L} \left[(V_G - V_T) V_D - \frac{V_D^2}{2} \right]$$

2. Saturation region:

$$I_D = \mu_n C_i \frac{Z_{eff}}{2L} \left[(V_G - V_T)^2 - 8r \left(\frac{k_B T}{q} \right) e^{\left(\frac{q(V_G - V_0 - V_D)}{k_B T} \right)} \right]$$

Note that the saturation equation above is the complete equation and if we take the long semiconductor assumption the second term can be ignored, in which case the equation essentially becomes identical to the one for the MOSFET. By looking at the equations, for a MOSFET and FinFET with identical gate lengths and material properties, due to the extra surface area provided by the fins, there is a larger inversion area, allowing for the diffusion of more carriers, increasing the flow of drain current [9].

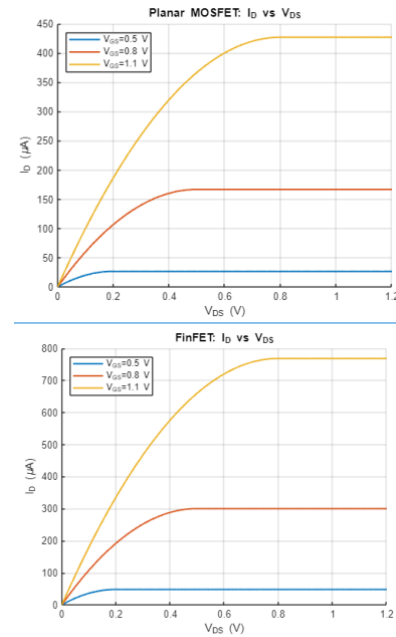


Figure 2: Difference in drain current between a n-channel MOSFET and n-channel FinFET at different gate voltages using the complete equations simulated on MATLAB with identical gate lengths and widths.

From figure 2 it is evident the significant impact the fins have in boosting the current driven through the FET. To further improve the performance additional fins can be added and the effect it would have is equivalent to multiplying the Z_{eff} by the N fins that are added.

FINFET VS. GAAFETS

While FinFETs are significant improvements to traditional planar MOSFETs, they have their limitations. In recent years as companies began shifting towards sub-5nm transistor architecture, even FinFETs begin to approach their limits of effective gate control. For this reason, the industry

is thus moving to a new architecture: the Gate-All-Around FET (GAAFET), also known as nanosheet or nanowire transistors. GAAFETs take the multi-gate concept to its ultimate form by completely surrounding the semiconductor channel with the gate electrode on all sides, rather than the three-sided gate of a FinFET [10].

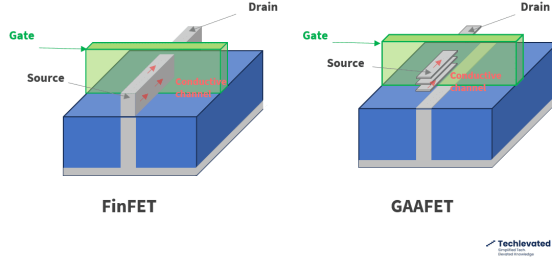


Figure 3: FinFET design vs GAAFET [11].

This can be explained through the electrostatic screening length, λ , which is the characteristic length that describes how short the channel length can be before short-channel effect takes over. The equations for scaling length are given below:

$$\lambda_{FinFET} \approx \sqrt{\frac{\epsilon_{si} t_{si}}{3\epsilon_{ox}}}$$

$$\lambda_{GAAFET} \approx \sqrt{\frac{2\epsilon_{si} t_{si}^2 \ln\left(1 + \frac{2t_{ox}}{t_{si}}\right) + \epsilon_{ox} t_{ox}^2}{16\epsilon_{ox}}}$$

Where ϵ_{si} , ϵ_{ox} are the permittivity's of silicon (channel) and gate oxide, t_{si} is the channel thickness, and t_{ox} is the oxide thickness. Simply computing the screening length with the same permittivity and channel thickness value will result in a smaller λ for the GAAFET [12].

That tighter screening also clarifies why GAAFET nanosheets outperform FinFETs when you try to increase drive current. In a FinFET the simple way to get more current is to make the fin wider or stack more fins, but you can only make the fins so wide before the FET becomes too large. GAAFETs turn this limitation into a strength because their λ is already intrinsically smaller, each nanosheet can be made wider in the horizontal direction while still remaining fully depleted across its thickness. Manufacturers can then stack several such wide, well-controlled sheets vertically inside one device, giving a much larger effective channel width, and hence a higher on-current, all within the same footprint [13].

Despite GAAFETs being an upgrade to FinFETs, FinFETs have not entirely been abandoned as they are still more affordable to manufacture, thus they are still being used in circuits that do not necessarily need the smallest transistor. Nonetheless, the superior performance of GAAFETs, and the promising results they have to offer, may eventually make them more affordable as they become more widespread within IC circuitry.

DEVICE APPLICATIONS

FinFET technology is broadly used across almost all semiconductor applications that demand high performance, low power, or high circuit density. Some examples of their applications include:

Digital Logic and Processors: Virtually all advanced CPUs, GPUs, and SoCs utilize FinFET transistors of 16 nm, 10 nm, 7 nm, 5 nm, etc. for their logic gates. High-profile processors from Intel, AMD, Apple, and Qualcomm are built with FinFETs, due to their high switching speed and power efficiency. For example, mobile SoCs in smartphones and tablets widely use 7 nm and 5 nm FinFETs to be able to provide billions of transistors for computing, at very low power consumption. This enables our smartphones to do almost anything without requiring frequent charging.

Memory (SRAM and DRAM): On-die static memory like SRAM (used in CPU/GPU caches) is implemented with FinFET devices. The 6T SRAM bit-cells benefit from FinFET's leakage reduction compared to planar MOSFETs [14]. FinFETs have also been explored in DRAM array access transistors and non-volatile memory periphery circuits. In high-density memory like SRAM, specialized FinFET device options have been developed with to have especially high threshold voltage, and low leakage to optimize memory stability and leakage.

Analog and Mixed-Signal: Although FinFETs were originally intended for digital logic, they are also used in analog/RF circuits. Analog-to-digital converters (ADCs), radio frequency (RF) transceivers, and power management circuits in modern mobile and communications chips may likely rely FinFET processes [10]. FinFET's intrinsically high gain and transconductance can be advantageous for analog performance.

High-Performance Computing (HPC) and AI: Data center accelerators, FPGAs, and AI chips rely on FinFETs to meet extreme performance demands. GPUs by Nvidia and AMD, AI accelerators like Google's TPU, and FPGAs by Xilinx/Intel all use FinFET transistor technology to pack billions of transistors while controlling power. FinFET's ability to provide high drive current makes it ideal for large, power-hungry chips such as those used in HPC and machine learning training [15].

In summary, any application that migrated to <20 nm technologies has inherently adopted FinFETs. The advantages of FinFETs in speed, power efficiency, and density have made them prevalent from consumer mobile devices to enterprise servers and specialized electronics. Even fields like medical imaging have benefited FinFETs serve as a versatile platform for both digital and analog needs in the semiconductor industry's most demanding applications.

INDUSTRIAL USE

Multiple industry leaders have successfully implemented FinFET technology in their products and fabrication processes.

Intel was the FinFET pioneer in industry implementation. Its 3rd Generation Core processors, codenamed Ivy Bridge, launched in 2012 were manufactured on Intel's 22 nm process, the first to use 3D Tri-Gate FinFETs in high-volume production [1]. Intel reported that these FinFETs provided ~37% performance gain at low voltage or 50% power reduction over 32 nm planar transistors [3]. The fins in Intel's 22 nm devices were approximately 8 nm thick at the top, 18 nm at the base, and 35 nm tall. Intel continued with FinFET transistors for subsequent nodes – 14 nm (Broadwell/Skylake CPUs), 10 nm (Ice Lake), and 7 nm (Intel 4). FinFETs have powered Intel's entire product line from high-end Xeon server CPUs to energy-efficient mobile chips since 2012. Intel's success demonstrated FinFET reliability at scale, and by 2015, virtually all mainstream CPUs used FinFETs [1].

TSMC Co. introduced FinFETs in its factory processes at the 16 nm node, which was quickly followed by 10 nm and a 7 nm generation in 2018 [16]. Fabless companies like AMD leveraged TSMC's FinFET processes to regain competitiveness. AMD's Zen microarchitecture which was first launched in 2017 on GlobalFoundries 14 nm FinFET, and later on TSMC 7 nm FinFET for Zen 2 in 2019 delivered huge performance-per-watt gains, largely thanks to FinFET efficiency. AMD's Radeon GPUs also shifted from 28 nm planar to 14/16 nm FinFET in 2016, enabling higher clock speeds and much lower power than the previous generation. Today, AMD's latest Ryzen CPUs and Radeon GPUs are built on 5 nm FinFET technology, containing tens of billions of FinFET transistors for enormous computing power [17].

Nvidia adopted FinFET-based processes for its GPU chips starting with the Pascal generation in 2016 which is built on TSMC's 16FF process [18]. This decision to shift from 28 nm planar to 16 nm FinFET allowed Nvidia to pack more cores and achieve ~2X performance per watt improvement on the GTX 10-series GPUs. Subsequent Nvidia architectures Volta, Turing, Ampere, and the latest Ada Lovelace have all utilized FinFET transistors from TSMC or Samsung foundries. For instance, the A100

datacenter GPU integrates 54 billion 7nm transistors, a feat made possible by using FinFET's density and power benefits.

Virtually every major chip manufacturer has FinFET experience. Apple designs its custom iPhone/iPad SOCs on FinFET. Qualcomm Snapdragon mobile processors moved to FinFET at 14 nm, 10 nm and beyond. Even IBM, though exiting commercial fab, demonstrated FinFETs in its research alliance. With so many industry leaders adopting the technology for its benefits, it underscores why FinFET became an industry-wide standard for leading-edge semiconductor technology in the 2010s [19].

MARKET ANALYSIS

The rise of FinFETs has translated into a massive and still-growing semiconductor market segment. FinFET technology is now fundamental to the production of advanced chips, which in turn drive the market for foundry services, EDA tools, and materials. As of mid-2020s, the global FinFET technology market is valued in the tens of billions of dollars. One industry analysis estimated the market at US\$27.2 billion in 2022, with a projected compound annual growth rate (CAGR) of ~22.5% from 2023 to 2031, reaching approximately US\$163.6 billion by 2031 [20]. This can be attributed to the adoption of FinFETs in 5G smartphones, high-performance computing, and AI accelerators. This is further justified by a separate Global Market Insights report projects a high-teens CAGR for FinFET 3D transistors through 2034, driven by GPUs, 7 nm/5 nm nodes and AI accelerators, corroborating the growth trajectory reported by Transparency Market Research [21]. The rapid expansion of data centers and demand for powerful processors in cloud computing has been a major factor fueling FinFET-based chip production, contributing to a market growth in excess of 20% CAGR in recent years. Several trends explain this strong market trajectory. First, FinFET is the default transistor architecture for any new leading-edge chip design, which means companies developing CPUs, GPUs, or ASICs must invest in FinFET-based process technology. This creates a stable pipeline of demand for FinFET wafer production at factories owned by companies like TSMC and Samsung. Second, FinFETs have enabled growth in emerging applications. For example, the rollout of 5G networks and advanced smartphones has increased demand for 7 nm and 5 nm FinFET chips in modems and application processors. Automotive electronics is another fast-growing sector: FinFET-based AI chips for vehicles are increasingly used, as they can deliver high performance under power and thermal constraints. In terms of market players, TSMC and Samsung Foundry, which manufacture FinFET chips for many clients, account for a large share of the FinFET market revenue. Intel also contributes significantly with its internal production for CPUs. Companies like Qualcomm, Apple, Nvidia, AMD indirectly drive FinFET wafer volume through their foundry orders.

In summary, the FinFET market is on the order of tens of billions of dollars annually and growing, underscoring the profound impact FinFETs have had on the semiconductor industry's economics and technology roadmap.

CONCLUSION

FinFETs have proven to be a pivotal innovation in semiconductor device engineering, prolonging Moore's Law at a time when planar transistor scaling was no longer thriving. Since their introduction in the late 1990s and early 2000s, FinFETs transitioned from research prototypes to the dominant transistor structure in virtually all high-performance and mobile chips. By leveraging a 3D fin-shaped channel with gates on multiple sides, FinFETs are able to better achieve control over short-channel effects, enabling transistors as small as 5 nm (gate length) to switch reliably with low leakage. From being a concept in a laboratory over 2 decades ago, to becoming the industry standard in the last decade shows the usefulness of the device. The device which operates as an enhancement-mode FET device, is able to achieve very high gain, control, and current flow while using similar footprints to a MOSFET. Its adoption in companies like Intel, AMD, Nvidia, and others illustrate that FinFETs have been a foundation for the computing advances in the past decade. Market analysis confirms that FinFET-based processes are a multi-billion-dollar industry that continues to grow as demand for advanced chips increases. However, it is evident that FinFETs are approaching their limits and GAAFETs will likely begin guiding the next generation of transistor architectures. FinFETs will nonetheless remain a critical part of semiconductor technology for the foreseeable future, as it enabled the era of true 3D transistor design.

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